

Data Science Project Proposal

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This project will examine healthcare needs and access across Oregon counties by analyzing the relationship between inequity indicators, health outcomes, and healthcare access capacity. The project seeks to identify counties where populations with elevated healthcare needs may face comparatively limited healthcare resources. A county-level analysis was chosen due to the prevalence of healthcare data only being available on that scale.

Data sources:

- Census Data
 - Population size
 - County area (square miles)
 - Race (calculated to be % of Non-White or Hispanic people (i.e. % POC))
 - Percent of People in Poverty
 - Percent of people over 65
 - Percent of people who own a vehicle
- Health Data
 - Health indicators (CDC/PLACES, 2023): [map](#) / [dataset](#)
 - Unmet need (OHSU, 2025): [map](#) / [dataset](#)
 - Also Urban v. Remote v. Rural
 - OHP Enrollment (OHA, 2026): [dashboard and dataset](#)
- Environmental Data
 - Air Quality
 - Air Quality and Health Outcomes Dashboard (OHA, continuously updated) - can't find downloadable data, may remove
 - Air Pollution by County (NIMHD, 2026): [table](#)
 - Broadband Access by County (NIMHD, 2026): [table](#)

Context on Health Care:

- Hospitals: [map](#) / [map](#)
 - Data on available hospital beds will be used to calculate the number of beds per capita and the number of beds per square mile.
 - Limitation: The layer is rather outdated and I haven't been able to find more up-to-date information.
- Oregon Rural Health, 2025: [report](#)
 - Hospital Types: [map](#) (pdf) / [definitions and table](#) (pdf)
 - Referral
 - Type a/b
 - DRG/Type c

Project Result:

Map 1: A Choropleth map of healthcare access will be evaluated using county-level healthcare infrastructure metrics, including:

- hospital beds per capita (size),
- hospital beds relative to county area (size),
- hospital service level availability (color).

Counties will also be colored according to whether urban/rural/remote on the map as well.

These will be mapped using the GIS layer and their data aggregate for each county profile. It will also be noted what the highest level of care available in each county is and the nearest county with the higher levels of care.

Profiles: You will be able to use the map(s) or a selector to pull up county-level data that displays a full county profile of the data for each category of data available.

To be added if I can figure out how to accomplish them:

Map 2: The following variables will be mapped either individually (or via a composite score if I can figure out how to make one) :

- OHP enrollment (%),
- Poverty (%),
- elderly population (% 65+),
- race/ethnicity (% POC),
- vehicle access (% w/o car),
- chronic illness (%),
- unmet healthcare need (2025 score),
- broadband access (%),
- and air quality burden (Micrograms per cubic meter (PM2.5)).

This will create a healthcare need index which will be mapped using a gradient scale.

Chart: If I am able to create sufficient need and healthcare availability scores, I would love to also include a “Access vs Need” Scatterplot.

- X-axis = hospital access score
- Y-axis = unmet healthcare need score
- Bubble size = population
- Color =urban/rural/remote